

KUNMING WUJIABA, China

WMO: 567780

Lat: 24.992N

Lon: 102.744E

Elev: 6217

StdP: 11.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	32.2	35.3	19.8	18.7	61.6	23.2	22.0	60.3	23.7	59.6	21.5	57.6	6.8	40	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.2	82.6	61.5	80.7	61.6	79.0	61.7	67.5	75.9	66.9	74.8	66.2	73.6	11.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	118.0	70.1	64.6	115.3	69.6	64.1	113.2	69.3	36.1	76.7	35.4	75.0	34.8	73.9	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.1	16.7	27.9	86.1	3.7	1.9	25.2	87.5	23.1	88.6	21.0	89.6	18.2	91.0		
(5)			26.2	70.0	3.6	1.6	23.6	71.1	21.5	72.0	19.5	72.9	16.9	74.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	49.7	53.4	59.0	64.6	67.4	69.3	69.1	68.8	66.1	61.7	55.3	49.7
	DBStd	8.53	4.63	5.78	5.48	5.15	5.14	3.73	2.73	2.61	4.03	4.54	4.45	4.92
	HDD50	172	55	32	10	1	0	0	0	0	0	1	11	62
	HDD65	2026	473	325	190	66	37	10	3	4	35	116	292	475
	CDD50	4265	47	127	290	438	539	580	593	582	482	365	169	53
	CDD65	644	0	0	6	53	111	140	132	121	66	15	0	0
	CDH74	2487	0	5	76	436	679	489	312	308	151	30	1	0
	CDH80	245	0	0	2	41	106	58	16	16	6	0	0	0
(14) Wind	WSAvg	7.1	7.6	8.8	9.3	9.0	8.0	6.9	5.8	5.4	5.5	6.0	6.4	6.7
(15) Precipitation	PrecAvg	38.6	0.6	0.5	0.7	1.1	3.3	6.9	8.3	7.8	4.7	3.3	1.4	0.6
	PrecMax	54.6	2.0	1.9	3.0	4.8	8.7	18.7	14.5	16.1	11.1	11.5	5.8	2.7
	PrecMin	22.3	0.0	0.0	0.0	0.0	0.3	1.4	2.9	2.2	1.1	0.4	0.0	0.0
	PrecStd	7.6	0.5	0.5	0.7	0.9	2.1	3.2	3.2	3.0	2.2	2.2	1.3	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.4	74.6	78.9	83.8	85.8	84.4	82.1	82.1	80.7	76.9	73.1	68.4
		MCWB	47.5	49.0	52.3	56.1	60.7	64.7	66.3	65.9	64.6	61.2	55.1	50.0
	2%	DB	66.2	71.5	75.9	80.8	82.7	81.6	79.2	79.2	77.6	74.6	69.6	65.7
		MCWB	47.4	48.9	51.8	55.7	60.0	64.5	66.0	65.3	63.9	60.4	54.0	48.9
	5%	DB	64.0	68.6	73.7	78.9	80.6	79.0	77.3	77.3	75.4	72.0	67.2	62.9
		MCWB	46.7	48.2	51.2	55.3	60.1	64.6	65.6	64.8	63.0	59.5	53.2	48.6
	10%	DB	61.1	66.3	71.5	76.7	78.4	76.9	75.3	75.3	73.4	69.9	64.7	60.5
		MCWB	45.9	47.7	50.9	55.0	60.0	64.5	65.2	64.5	62.4	59.3	52.6	47.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	52.9	55.9	60.7	66.4	68.3	68.9	68.3	67.2	64.2	60.0	54.4
		MCDB	61.2	65.9	68.8	74.5	76.2	77.2	76.3	77.5	76.1	72.4	65.6	61.3
	2%	WB	49.8	51.2	54.5	59.3	64.9	67.2	67.6	67.2	65.8	62.9	58.0	52.0
		MCDB	60.1	64.6	68.0	72.7	73.8	75.7	75.6	75.6	73.7	70.1	63.9	60.0
	5%	WB	48.5	49.9	53.4	58.1	63.8	66.4	66.9	66.4	65.0	62.0	56.3	50.4
		MCDB	59.1	63.9	67.3	71.4	72.4	74.4	74.3	73.9	72.0	68.5	62.6	58.8
	10%	WB	47.3	48.7	52.3	56.8	62.8	65.6	66.1	65.6	64.1	60.9	54.6	49.2
		MCDB	57.7	62.5	66.2	70.3	71.4	73.1	72.9	72.3	70.5	66.8	61.3	57.3
(35) Mean Daily Temperature Range	5% DB	MDBR	22.5	23.2	22.6	21.6	18.0	14.2	13.0	14.3	14.4	15.2	18.8	20.2
		MCDBR	26.8	27.2	26.1	24.4	22.3	18.0	16.4	17.5	17.5	18.7	22.3	24.6
		MCWBR	11.0	10.2	9.2	7.8	6.8	6.2	6.4	6.7	6.9	7.5	9.2	10.4
	5% WB	MCDBR	23.6	24.2	22.7	21.5	16.5	14.4	13.7	15.0	14.9	15.0	17.8	20.7
		MCWBR	10.9	10.2	9.0	8.1	6.7	5.9	6.1	6.4	6.5	6.9	8.8	9.9
(40) Clear-Sky Solar Irradiance	taub		0.336	0.350	0.500	0.491	0.424	0.408	0.422	0.460	0.439	0.417	0.370	0.361
	taud		2.365	2.334	1.871	1.967	2.268	2.354	2.300	2.164	2.206	2.229	2.300	2.285
	Ebn at Noon		291	295	254	260	277	279	275	265	269	269	276	276
	Edh at Noon		34	38	64	59	44	40	42	48	45	42	36	36
(44) All-Sky Solar Radiation	RadAvg		1218	1530	1644	1763	1671	1425	1364	1377	1223	1088	1192	1049
	RadStd		121	187	145	159	194	151	97	97	101	131	133	125

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-1.75	+1.15	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.76	N/A	N/A	+1.39	N/A	-0.40	N/A	N/A	+191	+108

Nomenclature: See separate page